

Let us understand this from the perspective of a car. To simulate a car in a difficult situation and examine how it will behave, we need to feed in some parameters. But the actual result is sometimes different from that obtained from the simulation. Instead, if we connect particular sensors, which in turn are getting directly displayed over the Digital Twin, then that will give a whole lot more realistic data.

Where Digital Twin meets AI

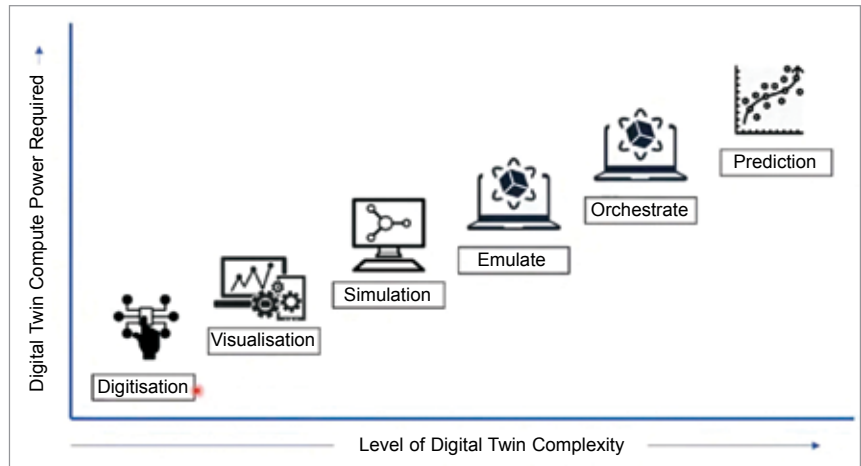
Digital Twin is a continuously learning system, powered by various algorithms.

Let us assume that we have several pumps in our factory. To make Digital Twin models of each pump, we first need to make 3D models of them. The engineer will work to collect the physical, mechanical, electrical, and other parameters to make a 3D design. And after that the actual objects will be fitted with sensors, which will deliver data to the digital replicas. And this Twin will display the live data and store it in a database (as historical data). This data can be used to train a model.

Let us now assume that hundreds of pumps are working and being tracked for the past one or two years. So, every time a pump gets damaged, some physical parameter gets changed. So, by having data, a model can be created that can predict any pump malfunctions. If it is repeating the same phenomena, which occurred earlier, a prediction can be made that this pump will not function well after some time. So, before the malfunction occurs, parts can be replaced or repaired, preventing any losses.

Jet engine case study

Suppose there is a jet engine that has sensors and is pushing data to its Digital Twin. If the jet engine has been running for some time, then chances are some damages might have occurred. Based on the previous historical database and the predictive model, Digital Twin will try to predict the estimated damage that has occurred in a particular part. But when the engine is physically inspected, there might not be the same scenario because predictive models are not 100% accurate.



Key parameters to consider when designing Digital Twin

Benefits to an organisation using Digital Twin

1. Digital Twin allows simulation and testing of new systems before manufacture, allowing results to be obtained beforehand.
2. Digital Twin improves productivity by predicting failure.
3. Digital Twin helps understand data to provide better service. If you have a multiple-threaded Digital Twin, there will be a lot of orchestration happening.
4. Digital Twin offers remote troubleshooting. It greatly increases visibility and traceability so that real-world events can be interpreted accurately.

That's a key point to understand while using Digital Twin. Therefore, to extract the actual damage that has happened, those references will need to be calculated once again and fed to the model to make it better. It is an iterative process. Slowly, the model will improve and its accuracy of prediction will become high.

This use case is also important for wind turbines. Digital Twin can predict the correct alignment of the wind turbine, that is, in which direction the wind turbine will get the most power from wind. Not only that, it can also tell at what time the predictive maintenance should happen.

Future

It is estimated that companies will be investing up to US\$1.1 trillion in IoT by 2023. There will be 1.9 billion 5G cellular subscriptions by 2024, which will push the use of IoT to a huge extent. The total economic impact of IoT could range anywhere between US\$4 trillion to US\$11 trillion by 2025.

According to Market & Markets, the global Digital Twin market was valued at US\$3.1 billion in 2020. But it is projected to reach US\$48.2 billion by 2026, which would be a huge

Models of Digital Twin

1. **Physics based model (Structural model):** Incorporates Newtonian mechanics. The sensor input is used for resolving forces and moments. Structural analysis/finite element analysis is used for resolving high-level structural state in every step. The decision is supported by aggregated performance analysis.
2. **Machine learning based model (AI based model):** Incorporates various computer science statistics and data science. With the assistance of a sensor, any unintended changes can be detected endlessly.

growth. And post-Covid that will increase even more rapidly.

In summary, the concept of Digital Twin has been there since long time. Now that we are noticing its potential and what we can achieve in collaboration with IoT and AI, it is gradually gaining acceptance by different industries. Digital Twin holds a big potential. Going by the numbers, it will soon find widespread applicability. **EFY**

The article is based on the talk Digital Twin & IoT by Arnab Ghosh, Developer and Researcher at Accenture Solutions, India that was presented at the February edition of Tech World Congress and India Electronics Week 2021.